

Julian Krese

julianwk123@outlook.com | Charlottesville, VA | (434) 566-3396

<https://www.linkedin.com/in/julian-krese/> | https://juliankrese.github.io/landing_page/home.html

EDUCATION

University of Virginia

Charlottesville, Virginia

Bachelor of Arts in Computer Science, Minor in Data Science, GPA: 3.963

May 2026

Coursework: Algorithms, Compilers, Computer Architecture, Computer Networks, Cybersecurity, Databases, Data Structures, Discrete Math, Linear Algebra, Machine Learning, Operating Systems, Software Development

EXPERIENCE

Climate Restoration Initiative - Environmental Institute (UVA)

Charlottesville, Virginia

Software Tool Developer

July 2024 - Present

- Collaborating with 5 schools at UVA to find economically profitable and sustainable solutions for carbon removal in the commonwealth of Virginia by looking into key geographic features on a county by county basis.
- Displaying and modifying GeoJSON files and data through Django and PostgreSQL for analysis.

Karpool

Cape Town, South Africa

Fullstack Software Engineering Intern

May 2024 - June 2024

- Built and integrated the payment system for a ride-sharing React-Native app, using the Paystack API and Azure.
- Designed high-fidelity Figma prototypes and wireframes, facilitating seamless collaboration across teams.

IXperience

Cape Town, South Africa

Software Engineering Student

May 2024 - June 2024

- Completed a 6 week software bootcamp taught by industry experts with an emphasis on full stack development.
- Learned industry standard tools and skills including but not limited to Bootstrap, React, and Node.

Department of Computer Science (UVA)

Charlottesville, Virginia

Undergraduate Teaching Assistant for CS 2100 and 3100

January 2024 - Present

- Providing help as a Teaching Assistant for Data Structures and Algorithms 1 and 2, taught in Java and Python.
- Teaching students Java and then the ability to implement and use data structures (i.e. trees, heaps, hash maps, etc.)
- Assisting with students' understanding of key concepts with an emphasis on algorithms related to Graphs, Divide and Conquer, Greedy, Dynamic Programming, Network Flow, and Machine Learning.

PROJECTS

Compiler - ANTLR, C++, LLVM

September 2024 - December 2024

- Fully extended the SIP language to TIP, a statically typed, imperative, and polymorphic language, including phases of parsing with ANTLR, semantic analyses, and code generation and optimizations with the LLVM IR.
- Added over 3,300 LOC and 358 lines of comments across 37 new files to the existing code base with 10,000 LOC

Blog - MongoDB, Express, React, Node, Bootstrap

May - June 2024

- Developed a MERN blog application that allows users to read and create blogs under authenticated accounts.
- Designed and implemented a comprehensive user interface featuring a secure login and registration page with JWT authentication, home page, categories page, author page, profile page, and blogs page.
- Used Google Cloud Buckets for secure storage.

Virtual Memory Allocation - C

March 2024

- Simulated OS assignment of physical memory to virtual memory
- Allocated and translated memory into multi-leveled page tables, with a configurable number of levels and size.

SKILLS & ACTIVITIES

Languages: C, CSS, HTML, Java, JavaScript, LLVM, Python, R

Libraries, Frameworks, and Other: ANTLR, Bootstrap, Express, Git, Keras, MongoDB, Node.js, NumPy, Pandas, React, SciKit-Learn, SQLite, Django, Azure, Google Cloud Platform, Web Development, Mobile App Development

Activities and Hobbies: Fly Fishing, Fly Tying, Golf, Hiking, Soccer, German